

4 Water is an important molecule in living organisms.

(a) Water is a dipolar molecule that forms hydrogen bonds with other water molecules.

(i) Explain the dipole nature of water.

(2)

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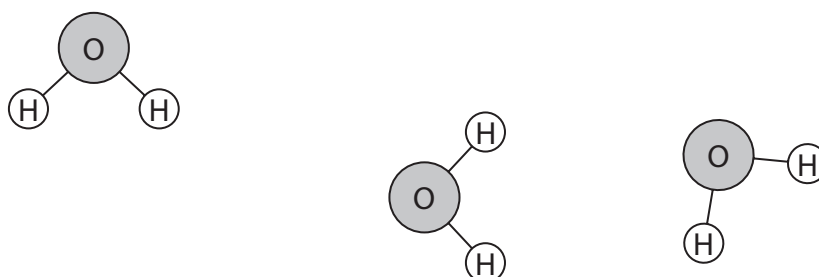
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(ii) The diagram shows three water molecules.

Complete the diagram to show a hydrogen bond between two water molecules.

(1)



(b) Water plays a role as a solvent.

Sodium chloride dissolves in water.

The diagram shows a sodium ion (Na^+).

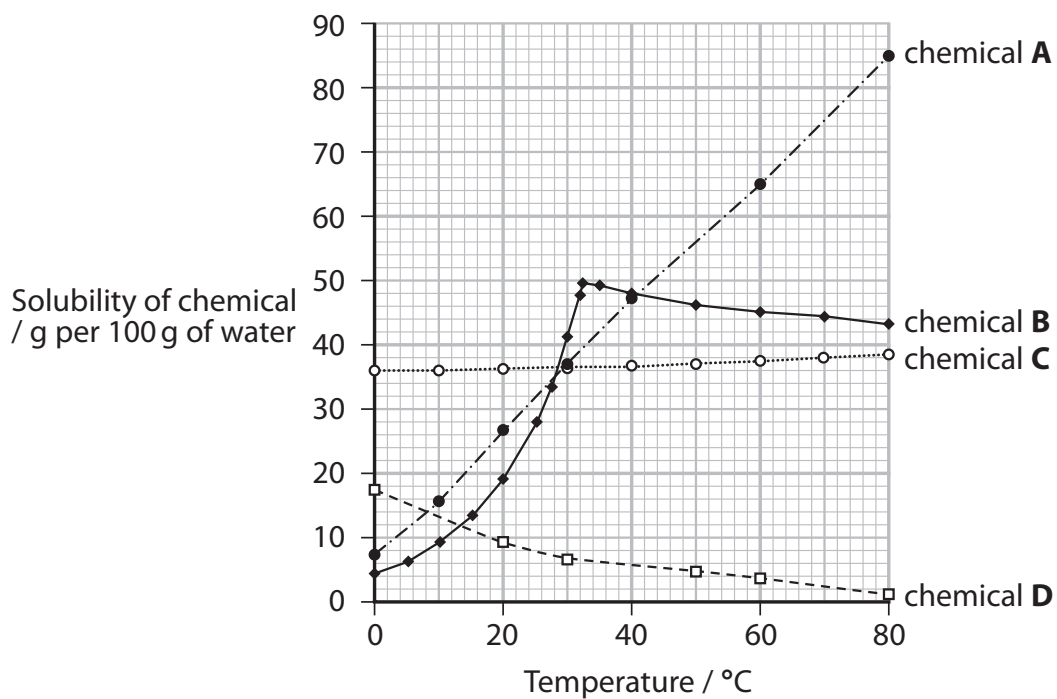
Complete the diagram to show the arrangement of two water molecules around a sodium ion when it is dissolved.

(1)



(c) Temperature affects the solubility of chemicals in water.

The graph shows the effect of temperature on the solubility of four chemicals, **A**, **B**, **C** and **D**.



Describe the effects of temperature on the solubility of these chemicals.

(4)

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(Total for Question 4 = 8 marks)

